

An LLM-Augmented Stacked Ensemble for Vietnamese Emotion Recognition on Social Media

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ABSTRACT

Fine-grained emotion recognition from Vietnamese social-media text is a difficult natural language processing problem due to the language’s low-resource status and severe class imbalance in available corpora. This paper proposes a two-stage framework evaluated on UIT-VSMEC, a seven-class Vietnamese social-media emotion corpus. In the first stage, we apply targeted data augmentation using Gemini-2.5-Flash via few-shot prompting, generating synthetic examples and paraphrasing existing ones for underrepresented emotion classes (Fear, Anger, Surprise). This raises CafeBERT’s weighted F1 from 62.8% to 67.1%. In the second stage, we construct a stacked ensemble of five heterogeneous base classifiers (PhoBERT, CafeBERT, ViBERT, Logistic Regression, LinearSVC), whose out-of-fold probability estimates are aggregated into structured meta-features and used to fine-tune a Gemma-2-9B-it meta-model with QLoRA (4-bit quantization). The full framework achieves 70.23% accuracy, 70.22% weighted F1, and 68.31% macro F1 on UIT-VSMEC, surpassing the prior CNN+word2vec weighted F1 of 59.74% and outperforming zero-shot Gemma prompting, published transformer baselines, and classical ensemble methods.

Index Terms—Vietnamese NLP, emotion recognition, data augmentation, large language models, ensemble learning, social media

I. INTRODUCTION

The rapid growth of Vietnamese social-media platforms—Facebook, TikTok, Zalo—has produced vast volumes of user-generated text carrying rich emotional signals. Mining these signals through fine-grained Emotion Recognition (ER) yields value for market research, opinion monitoring, and mental health screening, yet ER remains substantially harder than coarse sentiment polarity classification. Sentiment analysis on Vietnamese text—typically framed as a three-class task (positive, negative, neutral)—has benefited from larger annotated corpora, mature pre-trained language models, and relatively clear lexical polarity cues. Hybrid architectures such as PhoBERT combined with CNN-LSTM reach over 93% accuracy on the UIT-VSFC student-feedback corpus [1]–[3], and several deep-learning baselines report F1 scores above

90% on comparable polarity benchmarks. By contrast, fine-grained emotion recognition assigns each sentence to one of seven distinct affective states (e.g., Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, Other), where labels are linguistically subtler, more context-dependent, and frequently confounded in informal social-media writing [4]. On UIT-VSMEC, the best reported result from the original benchmark study reaches only 59.74% weighted F1 using CNN with word2vec embeddings [5]—a gap of more than 30 percentage points relative to polarity systems. Subsequent monolingual BERT models have narrowed but not closed this divide, indicating that scaling pre-training alone is insufficient for reliable fine-grained Vietnamese emotion classification.

Two interconnected obstacles drive this gap. First, Vietnamese is a low-resource language [6], [7]: annotated datasets are scarce, and existing corpora tend to be small. Second, UIT-VSMEC suffers from pronounced class imbalance: the dominant label *Enjoyment* constitutes roughly 28% of the training set, whereas minority emotions such as *Surprise* (~4.5%) and *Anger* (~6.9%) are severely underrepresented. Together, these constraints limit the robustness and generalization of individual classifiers.

We address both obstacles with a unified two-stage framework. In the first stage, we exploit the generative capabilities of a large language model (LLM), specifically Gemini-2.5-Flash [8], to perform targeted data augmentation for minority emotion classes using 10-shot prompting. In the second stage, we build a stacked ensemble whose base predictions are aggregated into structured meta-features and used to fine-tune a Gemma-2-9B-it meta-model with parameter-efficient QLoRA.

The main contributions of this paper are:

- 1) An LLM-based augmentation protocol that generates and paraphrases Vietnamese social-media sentences for minority emotion classes, demonstrably improving minority-class F1 scores with a controlled 10-shot prompting strategy.
- 2) A stacked ensemble framework that combines Vietnamese BERT models and classical TF-IDF classifiers with a QLoRA-fine-tuned Gemma meta-learner, achieving accuracy-weighted aggregation of diverse classifier signals.

- 3) Empirical validation on UIT-VSMEC demonstrating a new state-of-the-art of 70.23% accuracy, 70.22% weighted F1, and 68.31% macro F1, exceeding zero-shot LLM prompting, published BERT baselines, and prior ensemble methods.

II. RELATED WORK

A. Vietnamese Emotion Recognition

Ho et al. [5] introduced UIT-VSMEC, a 6,927-sentence seven-class corpus, establishing a CNN+word2vec baseline of 59.74% weighted F1. Nguyen and Nguyen [9] reached 64.40% through multi-label regression with preprocessing and key-clause extraction. Pre-trained transformers further narrowed the gap: PhoBERT [10] underpins hybrid CNN-LSTM architectures that achieve 93.24% accuracy on the polarity task [3], but fine-grained ER on informal social-media text remains harder. CafeBERT [11] (XLM-RoBERTa trained on Vietnamese Wikipedia and news) reports 66.12% F1 on VSMEC; ViSoBERT [12] reaches 68.37% weighted F1; and a GRU-CNN-BiLSTM-LSTM ensemble [13] attains 65.79% macro F1. Despite these advances, minority emotions such as Fear, Anger, and Surprise remain difficult to classify and leave room for data-centric and ensemble improvements.

B. LLM Augmentation and Ensemble Methods

Ding et al. [8] survey LLM-driven data creation and co-annotation, noting risks of contamination and model collapse. Ubani et al. [14] show zero-shot ChatGPT generation outperforms EDA and back-translation on SST-2 and SNIPS; Whitehouse et al. [15] extend LLM augmentation to cross-lingual settings; Üveges and Ring [16] confirm 10-shot LLM prompting yields macro-F1 gains for underrepresented emotion classes. However, prior work treats the LLM as an offline generator only, and prompt design for fine-grained Vietnamese emotion categories remains underexplored. On the ensemble side, Thin et al. [17] obtain 94.03% weighted F1 on VSFC [1] via soft-voting over five Vietnamese transformers; Niimi [18] unifies multiple LLM inferences through bagging and weighted averaging. Thin et al. [19] further evaluate zero- and few-shot LLM prompting for Vietnamese sentiment, while Zhang and Han [20] integrate commonsense knowledge into fine-grained aspect-level analysis, highlighting leakage risks with generative models. Our work extends these directions by using Gemini-2.5-Flash for targeted augmentation and as a QLoRA-fine-tuned meta-learner over heterogeneous ensemble signals.

III. METHODOLOGY

Fig. 1 illustrates the end-to-end pipeline. It consists of four stages: LLM data augmentation, text preprocessing, base classifier training with cross validation stacking, and meta-model fine-tuning.

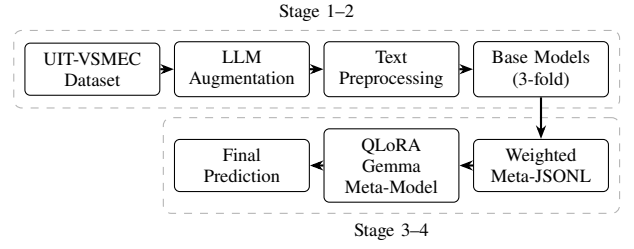


Fig. 1. Proposed pipeline: augmentation and preprocessing feed the base models; their cross validation probabilities form structured meta-features for QLoRA Gemma fine-tuning.

A. LLM Data Augmentation

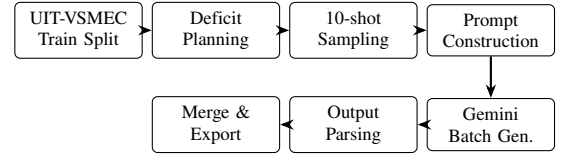


Fig. 2. Synthetic data creation pipeline for minority-class augmentation.

UIT-VSMEC exhibits a 6.4:1 imbalance between the most and least frequent classes. We address this by prompting Gemini-2.5-Flash [8] via the LangChain ChatGoogleGenerativeAI interface to generate additional training examples for the three most underrepresented emotions: Fear, Anger, and Surprise. Two complementary strategies are applied: (i) *generated*—the model synthesizes entirely new Vietnamese sentences conditioned on an emotion label and 10 randomly sampled in-context examples, creating novel scenarios not present in the original corpus; (ii) *paraphrased*—the model rewrites existing annotated sentences with linguistic variation while preserving the target emotion and original semantics. Each class is augmented to a target of 1,500 examples per strategy. Augmentation operates on the raw training split before any preprocessing. Fig. 2 shows the six-stage pipeline.

The pipeline operates as follows:

- 1) **Deficit planning.** For each minority class compute $\text{needed} = 1,500 - \text{current_count}$; classes already at the target are skipped.
- 2) **10-shot sampling.** Ten in-class sentences are sampled uniformly at random (with replacement when the class has fewer than ten examples) to form the few-shot context.
- 3) **Prompt construction.** A system message and structured human message are assembled from the selected template (generated or paraphrased).
- 4) **Batch generation.** Async calls to ChatGoogleGenerativeAI (Gemini-2.5-Flash, $T=0.9$, frequency penalty 0.5, concurrency 50) produce candidate sentences [21].
- 5) **Output parsing.** Lines matching `[n]. [sentence]` (example `[k]`) are extracted; malformed lines are discarded.

- 6) **Merge and export.** Accepted sentences are tagged (augmented=True, source), trimmed to the exact deficit, and concatenated with the original training split.

TABLE I

PROMPT DESIGN COMPARISON: GENERATED VS. PARAPHRASED STRATEGY

	Generated	Paraphrased
System role	VN social-media expert; express target emotion only	Gen-Z social-media expert; preserve meaning, tone, intent
Human task	Synthesize sentences {emotion} {emotion_vn}	NEW for / Rewrite sentences preserving SAME meaning and emotion
Key constraints	Natural VN style; length mix (5–10 / 10–20 / 20+ words); no other emotions	Semantic equivalence mandatory; no new facts or intensity; preserve negation and intensifiers
Output format	[n]. [sentence] (example [k])	

TABLE II

EXAMPLES OF AUGMENTED SENTENCES BY STRATEGY

Strategy	Original	Output	Emotion
Gen.	per không cho anh đi tập đầu nhà mất bỏ như chơi	Cái cảm giác bị bỏ rơi nó ám ảnh tui ghê lắm.	Fear
Gen.	sao mà mặt boss có thể gian phản diện đến thể nhỉ	Sao nhìn cái mặt của nhân vật phản diện nó gian manh vậy ta.	Surprise
Gen.	per bất hạnh thật tao với mày mua bánh ăn chung đi	Đời tôi khổ nạn lắm rồi, đừng ai làm phiền nữa.	Anger
Para.	sợ ngỗng còn hơn sợ chó :)))	Thà sợ chó còn hơn sợ ngỗng :)))	Fear
Para.	vì :) còn dứt tay túi áo ??	Trời đất ơi, còn dứt tay vào túi áo nữa hả?	Surprise
Para.	đẹp . đeo lưu luyến .	Thôi đi. Đéo cần vương vấn gì hết.	Anger

1) *Prompt Design.* Both strategies share the same LangChain ChatPromptTemplate structure: a *system message* that constrains style and emotion fidelity, and a *human message* that supplies the emotion label, its Vietnamese description, and 10 few-shot examples. Table I summarises the key differences; Table II shows representative outputs for both strategies. The shared output format requires each line to follow [n]. [sentence] (example [k]), enabling deterministic parsing.

B. Text Preprocessing

Vietnamese social-media text is noisy: users mix emojis, keyboard emoticons, slang, and short forms that carry strong

emotional meaning. If these signals are removed too early, models may lose useful cues. Following practices similar to prior Vietnamese social-media emotion work [6], [9], we therefore apply a shared cleaning pipeline to the training, validation, and test splits after augmentation.

Each sentence is cleaned in four steps. First, a curated dictionary of 116 emoji-to-Vietnamese mappings replaces Unicode emojis with short descriptive phrases (e.g., “vui vẻ” for happy faces and “buồn bã” for sad faces); unmatched emojis are then removed with a Unicode regex filter. Second, 39 text emoticons such as :) , :(, and T_T are mapped to Vietnamese words, with longer patterns matched first to avoid partial replacements. Third, 95 common internet abbreviations are expanded at the word level (e.g., *kolk/hok* → *không*, *dc/dc* → *được*). Fourth, URLs and HTML tags are stripped, non-Vietnamese characters are replaced with spaces while diacritics and basic punctuation are kept, repeated punctuation is collapsed, and whitespace is normalized.

After cleaning, duplicate sentences are removed so repeated content does not bias training. Stopwords are intentionally kept. Function words such as *không* (negation) and *rất* (intensifier) often change emotion meaning, and BERT-style encoders can use these tokens through self-attention [10], [11].

C. Base Classifiers

Five heterogeneous base classifiers are trained with stratified 3-fold cross-validation, each producing probability vectors across the seven emotion classes:

- **PhoBERT-v2** [10]: pre-trained on Vietnamese Wikipedia and news; fine-tuned for 20 epochs.
- **CafeBERT** [11]: a Vietnamese BERT variant from the VLUE benchmark; best single-model accuracy on VS-MEC.
- **ViBERT**: a Vietnamese BERT model (FPTAI/vibert-base-cased) further pre-trained on Vietnamese text.
- **Logistic Regression**: TF-IDF features, ℓ_2 regularization [22].
- **LinearSVC**: TF-IDF features, wrapped in CalibratedClassifierCV for probability calibration.

Each model is also refitted on the full training set to produce validation and test probability arrays. Classifier metrics are stored as JSON artifacts.

D. Stacked Ensemble with LLM Meta-Model

Weights and meta-features. Per-model weights are derived from validation accuracy:

$$w_i = \frac{\text{acc}_i}{\sum_j \text{acc}_j}. \quad (1)$$

A structured meta-prompt is assembled for each example, concatenating the raw input sentence, the five probability vectors, and the weighted-average soft prediction. This `compact_v1` representation encodes both individual model confidence and inter-model disagreement patterns.

QLoRA fine-tuning. Gemma-2-9B-it serves as the meta-learner. The model is fine-tuned with QLoRA (4-bit NF4 quantization, rank-16 adapters) using completion-only cross-entropy loss on the target token sequence `<label>EMOTION</label>`. Training uses the meta-JSONL dataset built from the predictions (unbiased by construction), which mitigates information leakage into the meta-model [23]. At inference, the model generates the emotion label conditioned on the structured meta-prompt for each test example.

IV. EXPERIMENTS

A. Dataset and Splits

UIT-VSMEC [5] contains 6,927 human-annotated sentences drawn from Vietnamese social-media platforms, covering seven emotion labels: Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, and Other. We follow the standard 8:1:1 split protocol, yielding approximately 5,548 training, 686 validation, and 693 test sentences after preprocessing. The training set is augmented to 1,500 examples per class for the minority classes before preprocessing.

B. Baselines and Evaluation

We evaluate two experimental conditions:

Augmentation ablation. CafeBERT is trained on three versions of the training data—original, paraphrased-augmented, and generated-augmented—and evaluated on the held-out test set using accuracy, weighted F1, and per-class F1 for the three minority emotions (Fear, Anger, Surprise).

Ensemble comparison. Table V compares the proposed stacked ensemble against (i) Gemma-2-9B-it zero-shot prompting on the same structured meta-prompts; (ii) published monolingual and multilingual transformer results on UIT-VSMEC (CafeBERT and XLM-RoBERTa-Large from VLUE [11], ViSoBERT [12]); (iii) the MLR pipeline with preprocessing and key-clause extraction [9]; and (iv) the GRU-CNN-BiLSTM-LSTM neural ensemble [13]. For context, the original UIT-VSMEC benchmark CNN+word2vec baseline [5] reached 59.74% weighted F1 but is omitted from Table V because it predates the transformer-centric comparison protocol used in our evaluation.

Evaluation metrics are accuracy, weighted F1, and macro F1.

V. RESULTS AND ANALYSIS

A. Impact of LLM Augmentation

Table III reports CafeBERT performance across the three training conditions.

TABLE III
IMPACT OF LLM AUGMENTATION ON CAFeBERT (UIT-VSMEC TEST SET)

Training Data	Acc.	W-F1	Fear	Anger	Surprise
Original	0.6344	0.628	0.67	0.38	0.60
Paraphrased	0.6676	0.6682	0.67	0.42	0.61
Generated	0.6703	0.671	0.68	0.46	0.61

Both augmentation strategies yield consistent improvements. The generated data condition produces the largest gains, raising accuracy from 63.44% to 67.03% and weighted F1 from 62.8% to 67.1%. Minority-class gains are particularly pronounced: Anger F1 rises from 38% to 46% (a gain of 8 percentage points), confirming that synthetic generation effectively addresses the most severely imbalanced classes. Fear and Surprise also improve, though more modestly, reflecting their lower initial imbalance. Paraphrasing yields intermediate improvements, suggesting that linguistic variation alone is beneficial but less impactful than generating entirely new examples. Table IV further analyses the quality of the generated sentences.

TABLE IV
QUALITY OF LLM-GENERATED TRAINING SENTENCES (5,010 SYNTHETIC VS. 5,548 ORIGINAL)

Metric	Original	Generated
Overall fidelity score	—	0.776
Distinct-2	0.593	0.451
Distinct-3	0.930	0.786
Mean sentence length (chars)	56.0	51.0
Train-test leakage (exact / near)	—	0 / 0
Emotion distribution KL div.	—	0.164

Fidelity = mean of stat, distribution, label, and correlation similarity.

Zero exact and near-duplicate test-set overlap confirms no evaluation leakage in the synthetic corpus. An overall fidelity score of 0.776 shows that generated text is statistically close to original social-media sentences; the KL divergence of 0.164 reflects intentional upsampling of minority classes, not a quality failure. Lower Distinct-2/3 values reveal moderate lexical repetition from 10-shot conditioning, yet generated data still outperforms paraphrasing because novel scenarios expand minority-class coverage more than surface variation alone.

B. Ensemble Performance

TABLE V
PERFORMANCE COMPARISON OF ENSEMBLE LEARNING METHODS AND BASELINES ON UIT-VSMEC (TEST SET). PUBLISHED TRANSFORMER ROWS FOLLOW VLUE/ViSoBERT BENCHMARK SETTINGS.

Method	Acc.	W-F1	M-F1
Proposed (stacked + QLoRA Gemma)	0.7023	0.7022	0.6831
Gemma zero-shot (ours)	0.6864	0.6809	—
MLR + preprocessing [9]	0.6436	0.6440	—
CafeBERT [11]	—	0.6612	—
ViSoBERT [12]	0.6810	0.6837	—
XLM-RoBERTa-Large [11]	0.6137	0.6020	—
GRU-CNN-BiLSTM-LSTM [13]	—	—	0.6579

W-F1: weighted F1; M-F1: macro F1; —: not reported.

Table V reports the full UIT-VSMEC benchmark comparison. The proposed framework achieves 70.23% accuracy and 70.22% weighted F1, establishing a new state-of-the-art and surpassing CNN+word2vec [5] by over 10 points.

The 2.13-point weighted-F1 gap over Gemma zero-shot confirms that QLoRA fine-tuning on ensemble-informed meta-

features yields a meaningful gain beyond zero-shot LLM inference. The best augmented single model (CafeBERT, 67.10% W-F1) trails the full ensemble by 3.1 points, showing that the meta-learner successfully integrates heterogeneous BERT and TF-IDF signals beyond any individual classifier.

VI. CONCLUSION

A. Summary

This paper proposed a two-stage framework for Vietnamese social-media emotion recognition: Gemini-2.5-Flash augmentation for minority classes, followed by a five-model stacked ensemble with a QLoRA-fine-tuned Gemma-2-9B-it meta-learner.

In the first stage, few-shot LLM augmentation consistently improved CafeBERT’s performance across all three under-represented emotion categories. Generated synthetic data—new sentences conditioned on 10 in-context examples—outperformed paraphrasing, raising weighted F1 from 62.8% to 67.1% and increasing Anger F1 by 8 percentage points (38% to 46%). A fidelity analysis confirmed that the synthetic corpus is statistically plausible (overall fidelity score 0.776, zero test-set leakage), establishing that new scenario coverage matters more than mere surface variation.

In the second stage, stacking heterogeneous BERT and TF-IDF signals through a QLoRA-fine-tuned meta-model produced a new state-of-the-art on UIT-VSMC: 70.23% accuracy, 70.22% weighted F1, and 68.31% macro F1, exceeding the original CNN+word2vec benchmark by over 10 percentage points [5]. The full ensemble surpassed zero-shot Gemma prompting by 2.13 weighted-F1 points and the best augmented single model (CafeBERT) by approximately 3.1 points, demonstrating that QLoRA meta-learning over structured ensemble signals yields meaningful gains beyond either component alone. Together, these results confirm that LLMs are effective both as offline data generators and as inference-time meta-learners in low-resource, class-imbalanced Vietnamese NLP.

B. Future Work

Future work will explore the following directions:

- 1) **Cross-dataset and cross-domain generalization.** Future work should evaluate the framework on related Vietnamese NLP tasks (aspect-level sentiment, hate-speech detection, multi-label corpora) to verify that LLM augmentation and stacked ensembling generalize beyond fine-grained emotion recognition.
- 2) **Automated quality validation for synthetic data.** The moderate Distinct-2/3 scores observed in Table IV call for automated filtering before synthetic examples enter training: candidate filters include perplexity thresholds, emotion-consistency scoring via a held-out classifier, semantic-similarity caps relative to near-duplicates, and LLM-as-judge fluency checks. The goal is to retain minority-class coverage gains while improving lexical diversity and reducing formulaic patterns.

- 3) **Dynamic, uncertainty-aware ensemble weighting.** Fixed validation-accuracy weights (Eq. 1) ignore per-example prediction confidence and inter-model disagreement. Replacing them with uncertainty-derived weights—computed from prediction entropy or calibrated confidence scores—may improve macro F1 on minority classes where static weights tend to favor majority-class-dominant models.
- 4) **Efficiency and deployability.** The current QLoRA meta-model requires substantial GPU memory at both training and inference time. Future work should investigate distilling the Gemma meta-learner into a smaller classifier or replacing it with a lightweight stacking head, making the framework practical for deployment in resource-constrained environments without sacrificing the gains from structured ensemble meta-features.

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